Title : Project Registration & Progress Review

Department: Computer Engineering	**Academic Year: 2026-27**
Semester : VII	**Group No. : BE - A32**
Project Title: An Explainable and Uncertainty-Aware AI Framework for Disaster Healthcare Demand Prediction and Multi-Objective Resource Allocation	
Project Area: Artificial Intelligence, Machine Learning & Operations Research	

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Project Synopsis

Project Title: An Explainable and Uncertainty-Aware AI Framework for Disaster Healthcare Demand Prediction and Multi-Objective Resource Allocation

1. Introduction

During large-scale disasters such as floods, earthquakes and cyclones, healthcare systems are often confronted with a sudden surge in patient load along with shortages of medicines, blood and other critical medical supplies. Resources also tend to be distributed unevenly across affected hospitals and regions, and decision-makers frequently struggle to judge which areas need support first. At present, emergency medical resources are largely allocated reactively, once a shortage has already been reported, rather than being planned proactively based on predicted demand. This project proposes an AI-based system that links disaster-impact prediction with healthcare-demand forecasting and resource-allocation optimisation, so that medicine and resource shortages can be anticipated and addressed before they become critical.

2. Problem Statement and Objectives

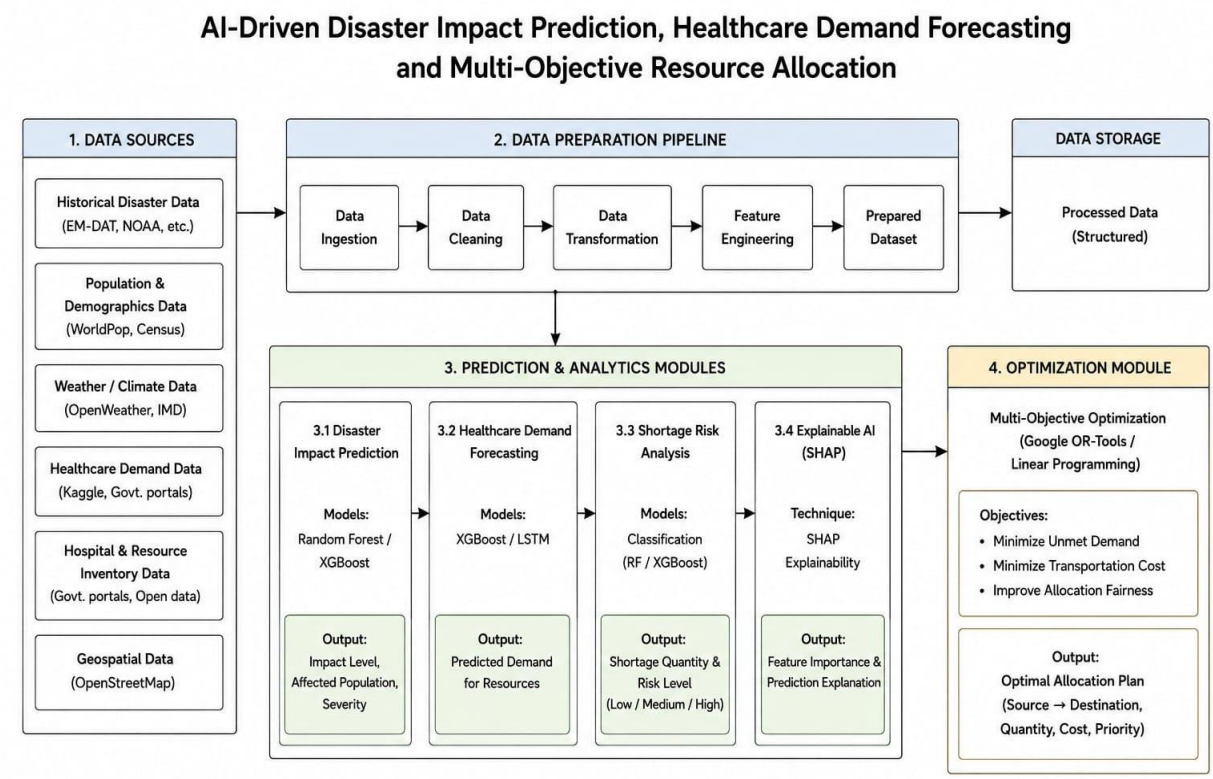
Problem Statement: Given historical disaster and healthcare data, design and evaluate an AI-based pipeline that predicts disaster impact, forecasts the resulting healthcare and medicine demand, identifies shortage risk in advance, and recommends an optimised allocation of limited medical resources across affected regions and hospitals.

Objectives:

- To predict the impact of a disaster (affected population, severity, likely injury/illness load) on nearby hospitals and regions using historical disaster records.
- To forecast short-term healthcare and medicine demand (e.g. antibiotics, ORS, IV fluids, pain medication, blood units) for the affected regions based on the predicted disaster impact.
- To build a shortage-risk early-warning module that flags medicines/resources likely to fall short of expected demand, along with a risk level (Low / Medium / High).
- To design an optimisation module that recommends redistribution of available stock across regions (e.g. transferring units from a low-risk region to a high-risk region) using optimisation/linear-programming techniques.
- To apply SHAP-based explainability to interpret the important factors influencing model predictions.

3. System Architecture

The proposed system follows a sequential pipeline in which each stage feeds the next:



1. Data Collection & Preprocessing
Collects disaster, population, healthcare-demand and resource/inventory data from verified public datasets. The data is cleaned, transformed and prepared for model training.
Input: Public datasets → **Output:** Clean, structured data
Technology: Python, Pandas, NumPy

2. Disaster Impact & Healthcare Demand Prediction
Uses historical disaster and healthcare data to estimate disaster impact and forecast the required quantity of selected medical resources.
Input: Disaster characteristics, affected population, historical demand → **Output:** Predicted impact and resource demand
Technology: XGBoost , Random Forest

3. Shortage Risk & Explainable AI
Compares predicted demand with available resources to identify shortage risk. SHAP explains which factors contributed to the prediction.
Input: Predicted demand + current inventory → **Output:** Shortage level/probability + explanation
Technology: Python, SHAP, ML models

4. Multi-Objective Resource Allocation
Uses optimization to determine how available resources should be redistributed between regions/hospitals while considering demand, supply, priority, distance and transportation constraints.
Input: Demand + inventory + constraints → **Output:** Optimal allocation plan
Technology: Google OR-Tools , Linear Programming

5. Decision-Support Dashboard & What-if Analysis
Displays predictions, shortage risks, explanations and recommended allocations. Users can modify disaster conditions or resource availability to simulate different scenarios.
Input: Model and optimization outputs → **Output:** Interactive recommendations and visualizations
Technology: Streamlit, Plotly/Folium
Overall flow
Public Data → Prediction → Shortage Risk + Explanation → Optimization → Decision

Environment:

The proposed system is primarily software-based and does not require dedicated hardware, sensors, or physical infrastructure. Therefore, its direct environmental impact is relatively low. By using publicly available datasets and cloud/local computing resources, the project avoids the environmental costs associated with manufacturing and deploying additional hardware.

The system can also contribute indirectly to environmental sustainability by enabling **better resource utilization and reducing unnecessary movement of medical supplies**. The optimization module can consider transportation distance and cost while recommending resource transfers between hospitals or regions. Efficient allocation can help reduce unnecessary transportation and avoid overstocking or wastage of medical resources.

The project will also consider computational efficiency by selecting appropriate machine-learning models and avoiding unnecessarily complex models when simpler models provide comparable performance.

Sustainability :

The project supports long-term sustainability by improving the way healthcare resources are planned and distributed during disaster situations. Instead of depending entirely on reactive decisions after shortages occur, the proposed system uses historical data and predictive models to support **Proactive Healthcare Planning**.

The system can be extended to different disaster types, geographical regions, healthcare resources and datasets without changing the basic architecture. This makes the proposed framework scalable and reusable.

The project primarily contributes to:

SDG 3 – Good Health and Well-being

The system supports early identification of healthcare-resource shortages and better preparedness for health emergencies. SDG 3 specifically includes strengthening capacity for health-risk reduction and management and improving access to essential healthcare and medicines.

SDG 11 – Sustainable Cities and Communities

The system supports disaster-resilient communities by helping authorities plan healthcare resources during disasters. SDG 11 includes reducing disaster impacts and strengthening disaster-risk reduction and resilience.

Safety and Security:

Because the system's recommendations directly influence emergency medical supply decisions, safety and cybersecurity are treated as first-class requirements. The system reports a confidence level with every shortage-risk prediction, flags predictions with low confidence for mandatory human review by hospital/disaster-management staff, and avoids automatically finalising a resource-transfer decision when confidence is below a defined threshold.

Project-specific safety considerations:

- Accurate, conservative shortage-risk classification that avoids under-estimating high-risk cases.
- Confidence score attached to every prediction and allocation recommendation.
- Mandatory human verification before any resource transfer is executed.
- Clear escalation/warning when a region is predicted to face a severe shortage.

Cybersecurity: the system should also address secure data ingestion from hospitals/disaster-management sources, authentication and role-based authorisation for hospital staff and administrators, secure APIs, database security and encryption of stored health and location data, and protection against malicious or malformed data uploads.

Ethical Considerations:

The project addresses ethical AI considerations including data privacy, informed consent for the use of hospital/patient-aggregate data, dataset bias (for example, under-representation of certain regions or disaster types in historical data), transparency and explainability of predictions, and human

verification of AI-generated allocation recommendations before they are acted upon. Ethics is particularly important here because incorrect or biased predictions could delay medical supplies to a vulnerable population.

Important considerations:

- Data privacy: disaster and hospital data may contain location, facility and aggregate patient information; only appropriately anonymised/aggregated data will be used, with clear data-retention limits.
- Bias mitigation: model performance will be checked across different regions and disaster types to avoid systematically under-serving any particular area.
- Explainability: SHAP-based explanations will be provided so that hospital administrators can understand why a region/resource was flagged as high risk.

Cost and Economic Feasibility :

The proposed project is economically feasible because it is primarily based on **open-source software, publicly available datasets and commonly available computing resources**.

Software

The project can use open-source technologies such as:

- Python
- Pandas
- NumPy
- Scikit-learn
- XGBoost
- SHAP
- OR-Tools

Therefore, there is no mandatory requirement for expensive proprietary software.

Data

The project is designed around **publicly available datasets**, reducing the cost associated with collecting proprietary disaster or healthcare data.

Hardware

No specialized hardware, sensors or embedded devices are required.

A standard computer capable of running Python and machine-learning models should be sufficient for the proposed prototype.

Economic benefits

The system could potentially reduce:

- unnecessary resource transfers
- excessive inventory
- resource wastage
- unmet healthcare demand
- inefficient emergency allocation
- decision-making time

The optimization component can also consider transportation cost when determining resource transfers.

Therefore, although the project itself is a student-level prototype, the underlying approach has potential economic value for hospitals, NGOs and disaster-management organizations.

Group No.	BE - A32		
Activity	Review Schedule	Progress Review Report submitted	Signature of Guide
Review 1	Mid Sem. Semester	Yes / No	
Review 2	End of Semester	Yes / No	

Format of Progress Review Report:

Review No.: 1	**Group No.: BE - A32**	**Date:**
Progress Review Report		
Signature of Guide:		

Review No.: 2	**Group No.: BE - A32**	**Date:**

Progress Review Report

Signature of Guide: